DDI CURRENT BOARD | PHASE 2-2

344 contract items + 1 ERC power flag / 7 INA228 channels / global-label sheet interfaces

Open a sheet below. Read README_KO.txt and verification before editing or releasing.

B1

12 V input / reverse protection / eFuse

B2

AUX 5 V buck - default ON

B3

VBAT 3.8 V buck - MCU enable / discharge

B4

CTRL 3.3 V LDO - default ON

B5_DIG

VDDI_DIG 1.24 V / 20 ohm / remote feedback

B5_VCI

VCI 3.0 V / 100 ohm / remote feedback

B5_VDDI

VDDI 1.8 V / 100 ohm / remote feedback

B6_IO

VDDIO 1.2 V - non-measurement rail

B6_SYS

SYS 1.8 V - non-measurement rail

B7_AVDDH1

AVDDH1 INA228 / address 0x43

B7_AVDDH2

AVDDH2 INA228 / address 0x44

B7_VCI

VCI INA228 / address 0x42

B7_VDDI

VDDI INA228 / address 0x41

B7_VDDI_DIG

VDDI_DIG INA228 / address 0x40

B7_VGH

VGH INA228 / address 0x45

B8_INA

VGL negative-rail current measurement

B8_ISO

VGL I2C isolation - ISO1641

B8_PWR

VGL floating power - RFM / 3.3 V LDO

B9

MOSCATO placeholder - NOT physical pinout

B10_A

Test points 1 / power and feedback sense

B10_B

Test points 2 / enable, PG, I2C and alerts

B11_CLOCK

8 MHz HSE / calibrated timebase

B11_CORE

STM32F072CBT6 / full 48-pin physical mapping

B11_DEBUG

SWD / reset / BOOT0

B11_POWER

MCU decoupling / VDDA filter

B12_CONNECTOR

USB-C host side / CC / ESD / shield

B12_ISOLATOR

ADuM4160 / full-speed isolation

B13

3.3 V to 1.8 V / MOSCATO EN and DDI reset

B14

DUT connectors / adjacent ground returns

B15

LEDs / mechanical / silkscreen / debug pads

RED FLAGS: MOSCATO physical pinout; custom footprints; input surge; low-current tare / temperature.

MCU / USB / level shifter / DUT headers included. PCB layout and physical release validation remain pending.

Engineering review draft - NOT production release

Sheet: /
File: DDI_Phase2_2.kicad_sch

Title: DDI Current Measurement Board / Phase 2-2

Size: A2	Date: 2026-10-11	**Rev: Schematic A**
KiCad E.D.A. 8.0.9		Id: 1/31

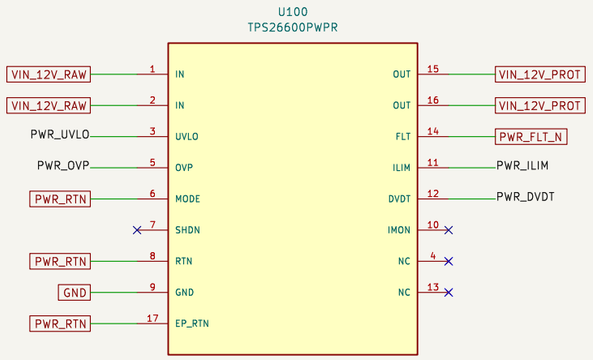
B1 / 12 V input / reverse protection / eFuse

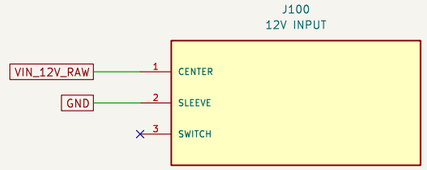
12 V input. UVLO ~9.922 V; OVP ~14.874 V; ILIM ~1.49 A.

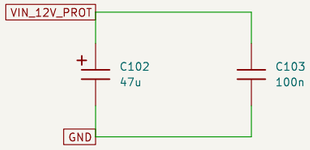
RTN is NOT GND. EP17 follows RTN. SHDN intentionally open.

PO: TVS clamp exceeds buck absolute maximum; validate surge shutdown.

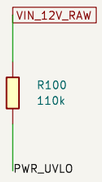
DC jack pin mapping and protection timing require hardware review.

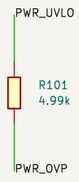


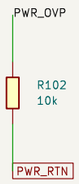


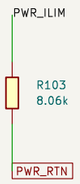


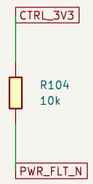










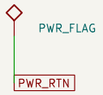












Same-name global labels connect between sheets. Local labels connect within this sheet only.

DNP components remain in the structural netlist; their population state is stored in the symbol.

REVIEW DRAFT — hardware validation pending

DDI Current Measurement Board

Sheet: /B1/

File: B1.kicad_sch

Title: 12 V input / reverse protection / eFuse

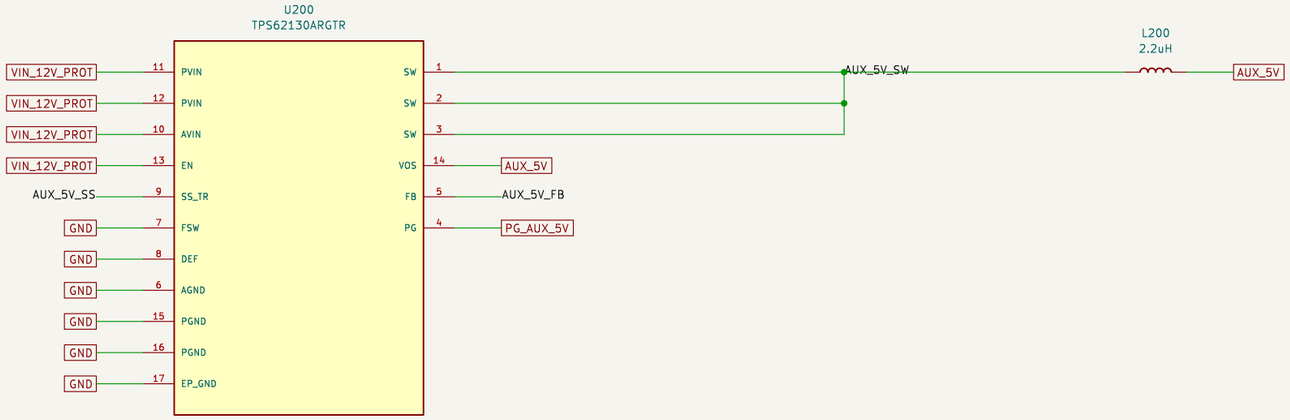
Size: A3	Date: 2026-10-11	**Rev: 2.2-A**
KiCad E.D.A. 8.0.9		Id: 2/31

B2 / AUX 5 V buck — default ON

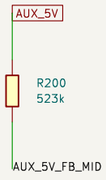
TPS62130A: 5.0 V / 2.2 uH / 2.5 MHz; default enabled.

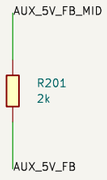
Input effective C >= 10 uF. Output effective C 22...44 uF.

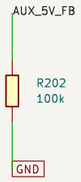
AVIN/PVIN loops and exposed pad return must be short.

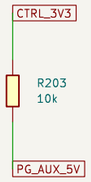
















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REVIEW DRAFT — hardware validation pending

DDI Current Measurement Board

Sheet: /B2/

File: B2.kicad_sch

Title: AUX 5 V buck — default ON

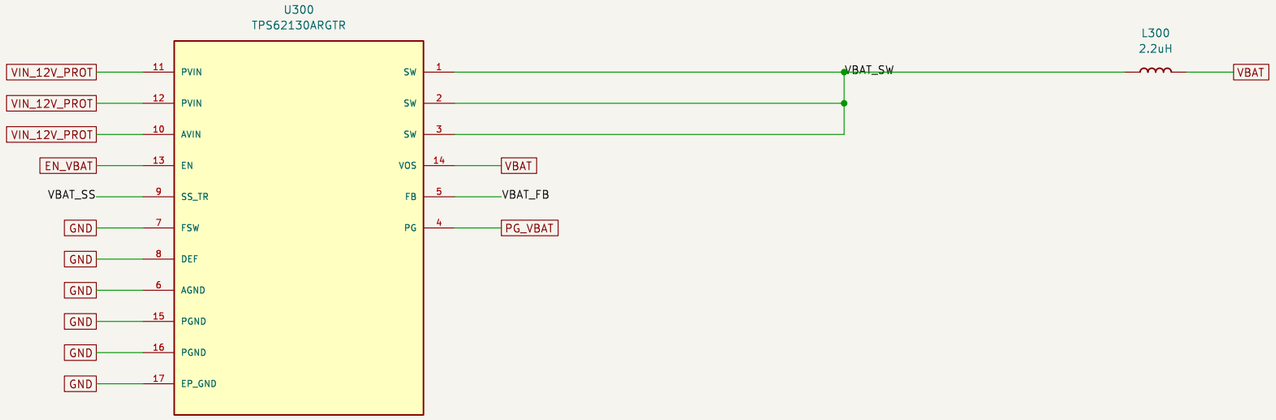
Size: A3	Date: 2026-10-11	**Rev: 2.2-A**
KiCad E.D.A. 8.0.9		Id: 3/31

B3 / VBAT 3.8 V buck — MCU enable / discharge

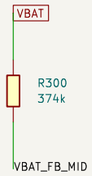
TPS62130A: 3.8 V. EN_VBAT has 100k pulldown.

1k bleeder: 3.8 mA; 14.44 mW; 44uF discharge to 0.3V ~112ms.

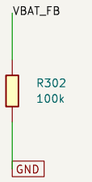
Total discharge time increases with MOSCATO input capacitance.

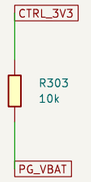


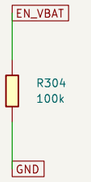


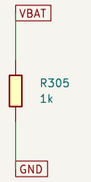
















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REVIEW DRAFT — hardware validation pending

DDI Current Measurement Board

Sheet: /B3/

File: B3.kicad_sch

Title: VBAT 3.8 V buck — MCU enable / discharge

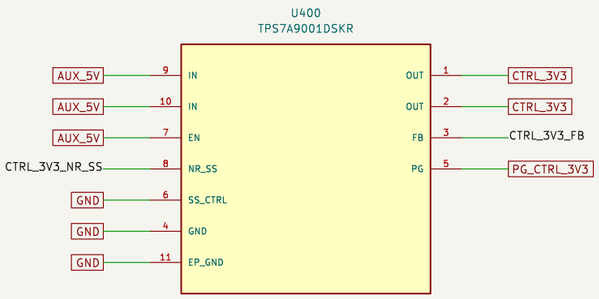
Size: A3	Date: 2026-10-11	**Rev: 2.2-A**
KiCad E.D.A. 8.0.9		Id: 4/31

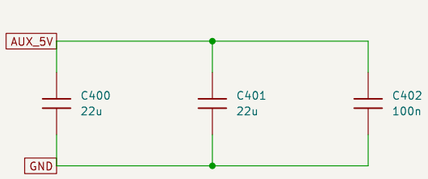
B4 / CTRL 3.3 V LDO — default ON

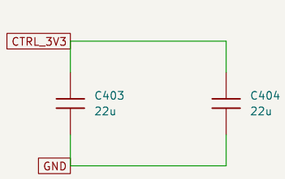
TPS7A90: 3.3 V, default enabled from AUX_5V.

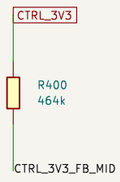
Shared CTRL_3V3 budget includes MCU, analog Iq and other LDO loads.

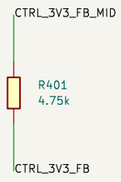
Accuracy below 5 mA and temperature drift require characterization.

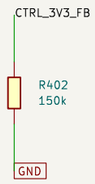


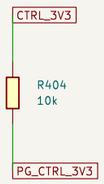


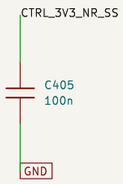














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REVIEW DRAFT — hardware validation pending

DDI Current Measurement Board

Sheet: /B4/

File: B4.kicad_sch

Title: CTRL 3.3 V LDO — default ON

Size: A3	Date: 2026-10-11	**Rev: 2.2-A**
KiCad E.D.A. 8.0.9		Id: 5/31

B5_DIG / VDDI_DIG 1.24 V / 20 ohm / remote feedback

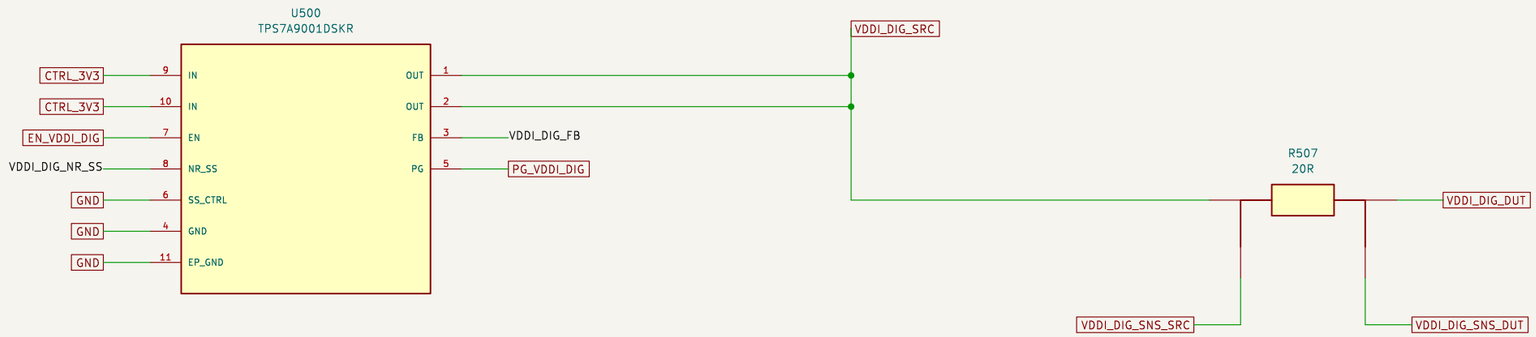
REMOTE jumper fitted / LOCAL jumper DNP: fit EXACTLY ONE.

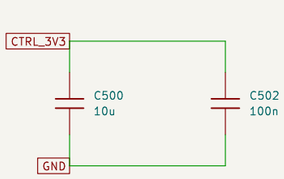
Fitting both bypasses the shunt. Fitting neither opens feedback.

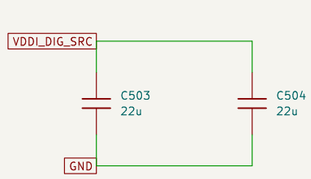
Shunt has 2 physical terminals / 4 logical Kelvin pads (custom PCB footprint pending).

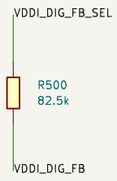
Divider load is included in measured current: tare calibration required.

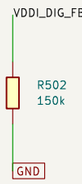
Low-load accuracy, load-removal overshoot and <=5 C local rise: bench validation.

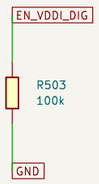


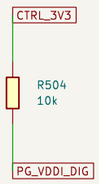


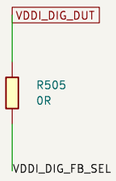




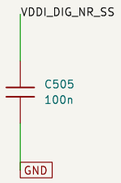














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REVIEW DRAFT — hardware validation pending

DDI Current Measurement Board

Sheet: /B5_DIG/

File: B5_DIG.kicad_sch

Title: VDDI_DIG 1.24 V / 20 ohm / remote feedback

Size: A3	Date: 2026-10-11	**Rev: 2.2-A**
KiCad E.D.A. 8.0.9		Id: 6/31

B5_VCI / VCI 3.0 V / 100 ohm / remote feedback

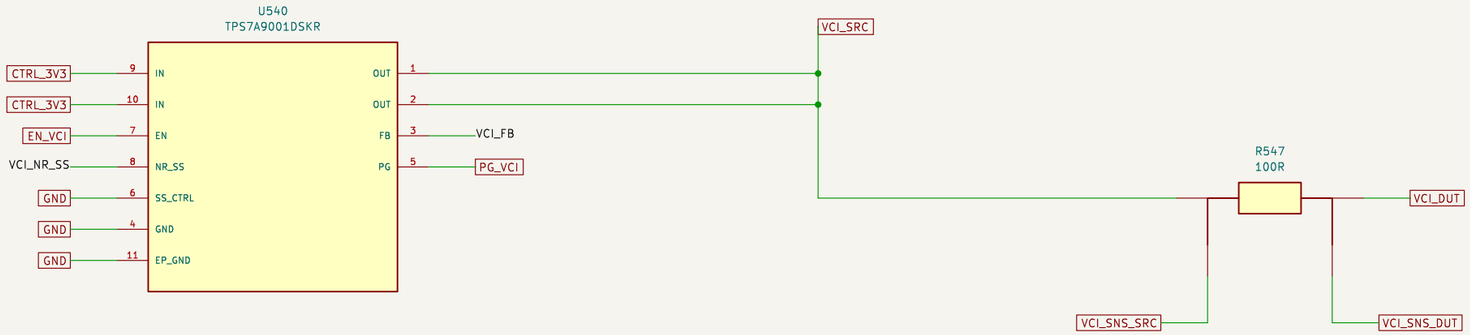
REMOTE jumper fitted / LOCAL jumper DNP: fit EXACTLY ONE.

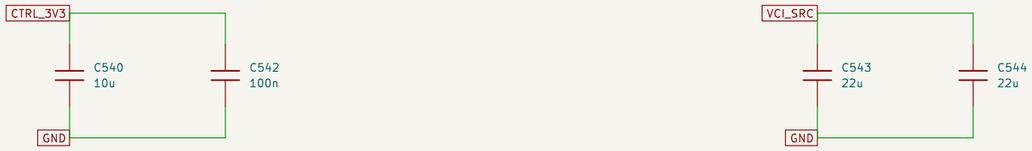
Fitting both bypasses the shunt. Fitting neither opens feedback.

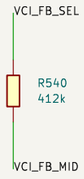
Shunt has 2 physical terminals / 4 logical Kelvin pads (custom PCB footprint pending).

Divider load is included in measured current: tare calibration required.

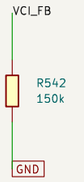
Low-load accuracy, load-removal overshoot and <=5 C local rise: bench validation.

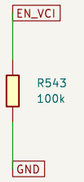




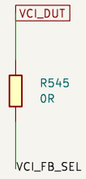


















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REVIEW DRAFT — hardware validation pending

DDI Current Measurement Board

Sheet: /B5_VCI/

File: B5_VCI.kicad_sch

Title: VCI 3.0 V / 100 ohm / remote feedback

Size: A3	Date: 2026-10-11	**Rev: 2.2-A**
KiCad E.D.A. 8.0.9		Id: 7/31

B5_VDDI / VDDI 1.8 V / 100 ohm / remote feedback

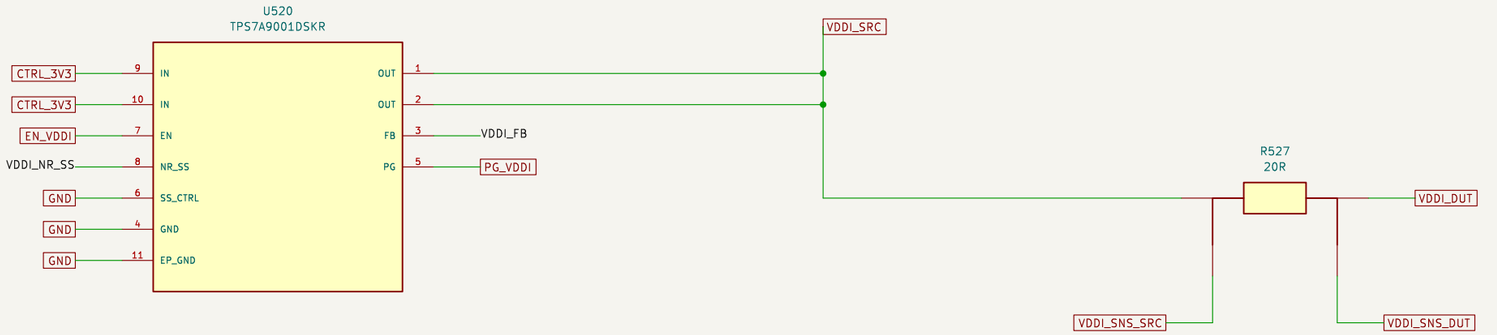
REMOTE jumper fitted / LOCAL jumper DNP: fit EXACTLY ONE.

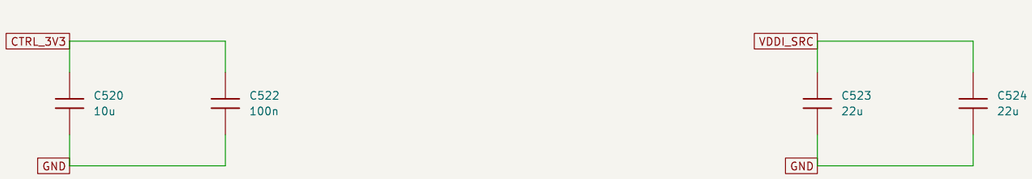
Fitting both bypasses the shunt. Fitting neither opens feedback.

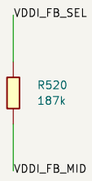
Shunt has 2 physical terminals / 4 logical Kelvin pads (custom PCB footprint pending).

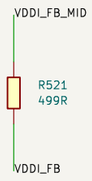
Divider load is included in measured current: tare calibration required.

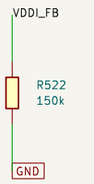
Low-load accuracy, load-removal overshoot and <=5 C local rise: bench validation.

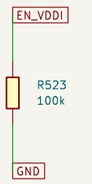


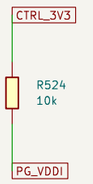


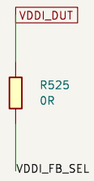


















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DNP components remain in the structural netlist; their population state is stored in the symbol.

REVIEW DRAFT — hardware validation pending

DDI Current Measurement Board

Sheet: /B5_VDDI/

File: B5_VDDI.kicad_sch

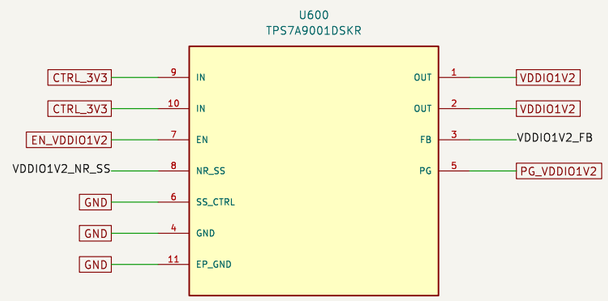
Title: VDDI 1.8 V / 100 ohm / remote feedback

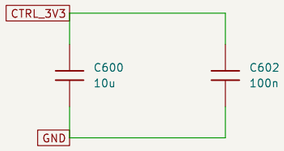
Size: A3	Date: 2026-10-11	**Rev: 2.2-A**
KiCad E.D.A. 8.0.9		Id: 8/31

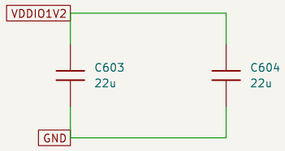
B6_IO / VDDIO 1.2 V — non-measurement rail

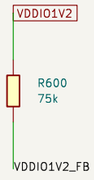
Non-measurement supply; never tie SYS_1V8 to measurement VDDI.

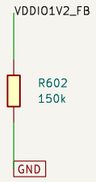
EN connected to MCU in Phase 2-2. Do not add duplicate PG pullups.

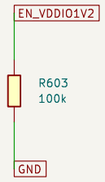


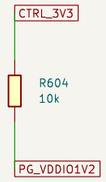


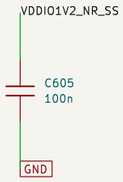


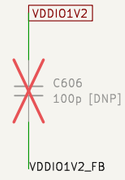












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REVIEW DRAFT — hardware validation pending

DDI Current Measurement Board

Sheet: /B6_IO/

File: B6_IO.kicad_sch

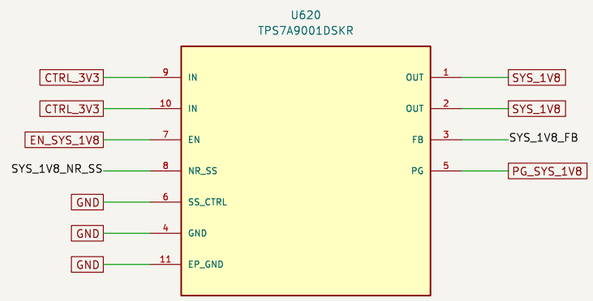
Title: VDDIO 1.2 V — non-measurement rail

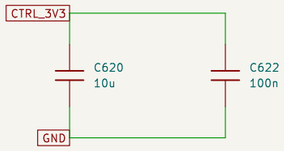
Size: A3	Date: 2026-10-11	**Rev: 2.2-A**
KiCad E.D.A. 8.0.9		Id: 9/31

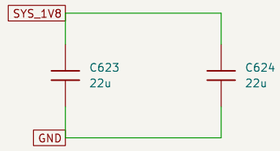
B6_SYS / SYS 1.8 V — non-measurement rail

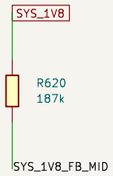
Non-measurement supply; never tie SYS_1V8 to measurement VDDI.

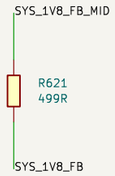
EN connected to MCU in Phase 2-2. Do not add duplicate PG pullups.

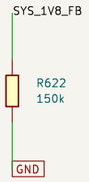


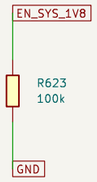


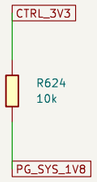
















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REVIEW DRAFT — hardware validation pending

DDI Current Measurement Board

Sheet: /B6_SYS/

File: B6_SYS.kicad_sch

Title: SYS 1.8 V — non-measurement rail

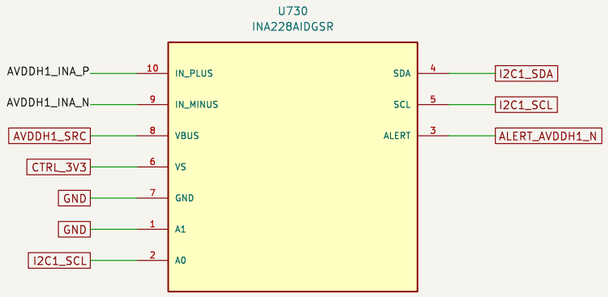
Size: A3	Date: 2026-10-11	**Rev: 2.2-A**
KiCad E.D.A. 8.0.9		Id: 10/31

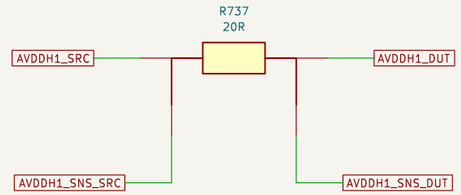
B7_AVDDH1 / AVDDH1 INA228 / address 0x43

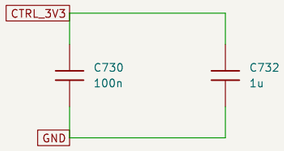
VBUS connects to source side of shunt. Kelvin sense wires remain separate.

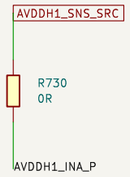
Input links: 0R fitted. Differential 1nF capacitor: DNP by default.

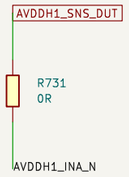
ALERT is open drain; pullup is included. MCU connection included in Phase 2–2.

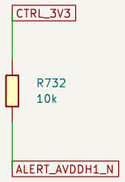


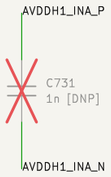












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REVIEW DRAFT – hardware validation pending

DDI Current Measurement Board

Sheet: /B7_AVDDH1/

File: B7_AVDDH1.kicad_sch

Title: AVDDH1 INA228 / address 0x43

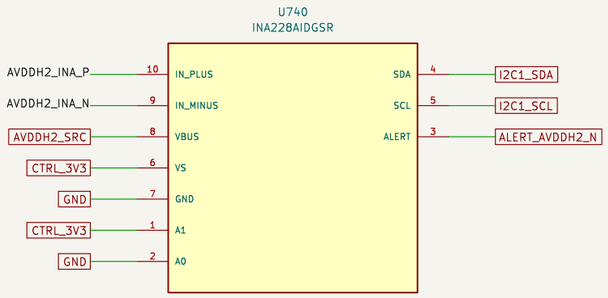
Size: A3	Date: 2026-10-11	**Rev: 2.2-A**
KiCad E.D.A. 8.0.9		Id: 11/31

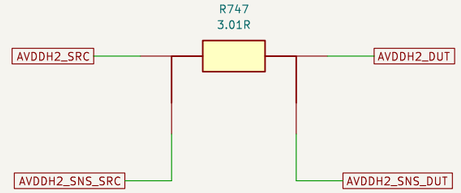
B7_AVDDH2 / AVDDH2 INA228 / address 0x44

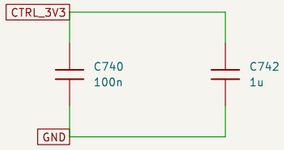
VBUS connects to source side of shunt. Kelvin sense wires remain separate.

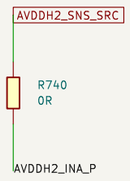
Input links: 0R fitted. Differential 1nF capacitor: DNP by default.

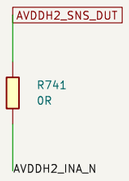
ALERT is open drain; pullup is included. MCU connection included in Phase 2–2.

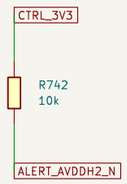


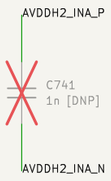












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REVIEW DRAFT — hardware validation pending

DDI Current Measurement Board

Sheet: /B7_AVDDH2/

File: B7_AVDDH2.kicad_sch

Title: AVDDH2 INA228 / address 0x44

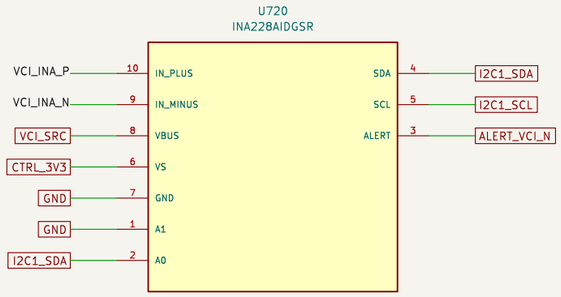
Size: A3	Date: 2026-10-11	**Rev: 2.2-A**
KiCad E.D.A. 8.0.9		Id: 12/31

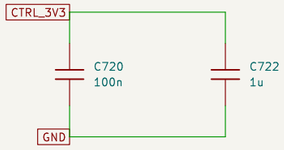
B7_VCI / VCI INA228 / address 0x42

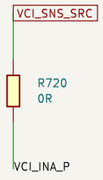
VBUS connects to source side of shunt. Kelvin sense wires remain separate.

Input links: 0R fitted. Differential 1nF capacitor: DNP by default.

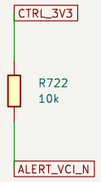
ALERT is open drain; pullup is included. MCU connection included in Phase 2–2.

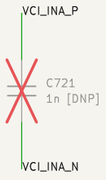












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REVIEW DRAFT – hardware validation pending

DDI Current Measurement Board

Sheet: /B7_VCI/

File: B7_VCI.kicad_sch

Title: VCI INA228 / address 0x42

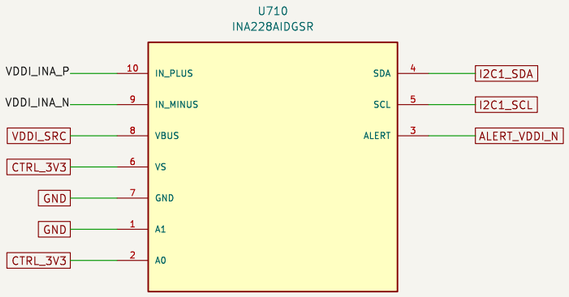
Size: A3	Date: 2026-10-11	**Rev: 2.2-A**
KiCad E.D.A. 8.0.9		Id: 13/31

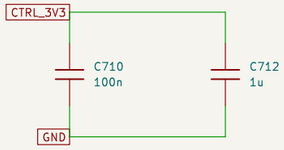
B7_VDDI / VDDI INA228 / address 0x41

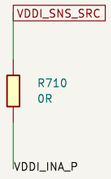
VBUS connects to source side of shunt. Kelvin sense wires remain separate.

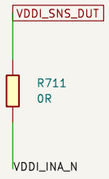
Input links: 0R fitted. Differential 1nF capacitor: DNP by default.

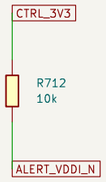
ALERT is open drain; pullup is included. MCU connection included in Phase 2–2.

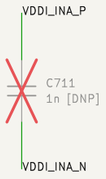












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REVIEW DRAFT — hardware validation pending

DDI Current Measurement Board

Sheet: /B7_VDDI/

File: B7_VDDI.kicad_sch

Title: VDDI INA228 / address 0x41

Size: A3	Date: 2026-10-11	**Rev: 2.2-A**
KiCad E.D.A. 8.0.9		Id: 14/31

B7_VDDI_DIG / VDDI_DIG INA228 / address 0x40

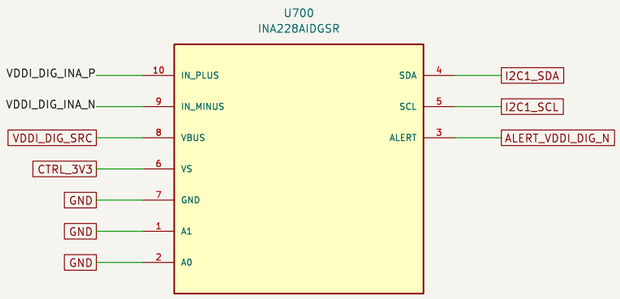
VBUS connects to source side of shunt. Kelvin sense wires remain separate.

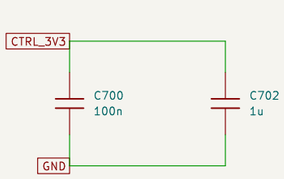
Input links: OR fitted. Differential 1nF capacitor: DNP by default.

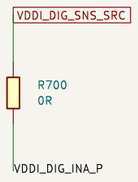
ALERT is open drain; pullup is included. MCU connection included in Phase 2–2.

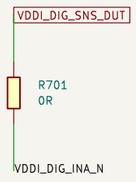
SOVL transparent ALERT / both-edge timer capture; ADC_CONFIG 0xAB28, DIAG upper bits 0000.

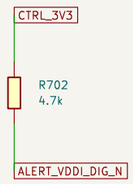
Raw 100uA threshold: 0x0190. 120Hz blank is shorter than ADC cadence; use asynchronous averaging.



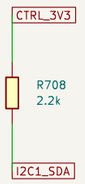














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REVIEW DRAFT – hardware validation pending

DDI Current Measurement Board

Sheet: /B7_VDDI_DIG/

File: B7_VDDI_DIG.kicad_sch

Title: VDDI_DIG INA228 / address 0x40

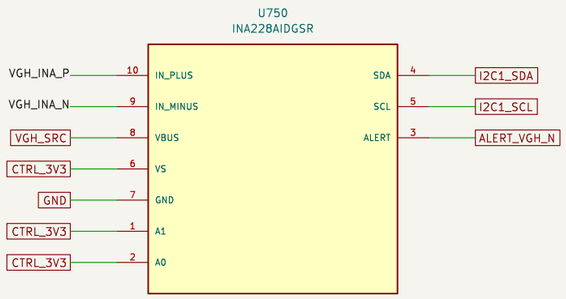
Size: A3	Date: 2026-10-11	**Rev: 2.2-A**
KiCad E.D.A. 8.0.9		Id: 15/31

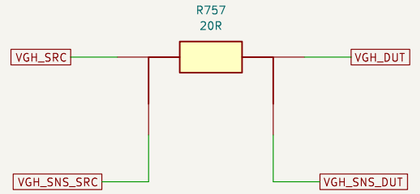
B7_VGH / VGH INA228 / address 0x45

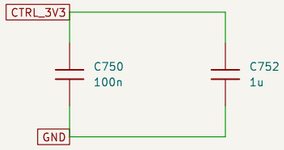
VBUS connects to source side of shunt. Kelvin sense wires remain separate.

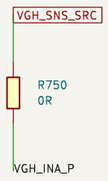
Input links: 0R fitted. Differential 1nF capacitor: DNP by default.

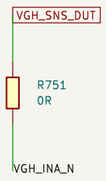
ALERT is open drain; pullup is included. MCU connection included in Phase 2–2.

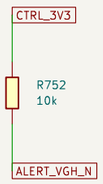














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REVIEW DRAFT – hardware validation pending

DDI Current Measurement Board

Sheet: /B7_VGH/

File: B7_VGH.kicad_sch

Title: VGH INA228 / address 0x45

Size: A3	Date: 2026-10-11	**Rev: 2.2-A**
KiCad E.D.A. 8.0.9		Id: 16/31

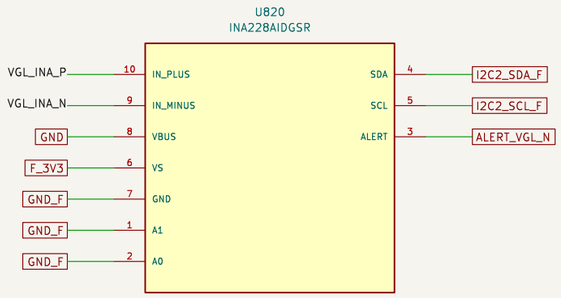
B8_INA / VGL negative-rail current measurement

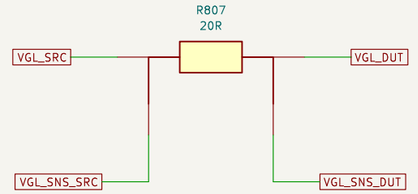
VGL polarity: IN+ = DUT side; IN− = SOURCE side; VBUS = system GND.

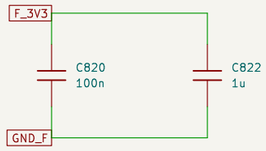
INA ground = GND_F = VGL_SRC via NT800. Address 0x40 on isolated I2C2.

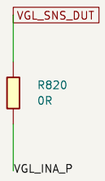
VGL_DUT = −VBUS + VSHUNT. Local ALERT is NOT a system GPIO.

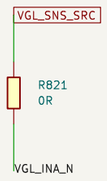
Kelvin pad pairs 1/3 and 2/4 are copper ties in the custom footprint.

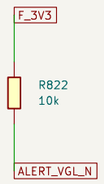


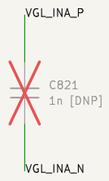












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REVIEW DRAFT — hardware validation pending

DDI Current Measurement Board

Sheet: /B8_INA/

File: B8_INA.kicad_sch

Title: VGL negative-rail current measurement

Size: A3	Date: 2026-10-11	**Rev: 2.2-A**
KiCad E.D.A. 8.0.9		Id: 17/31

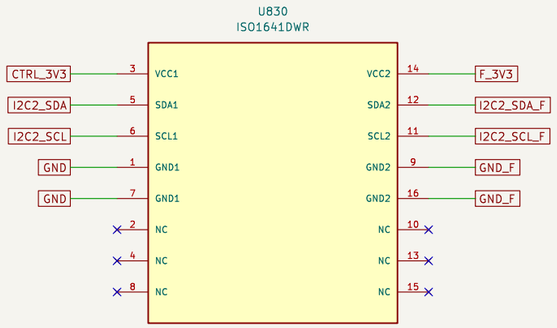
B8_ISO / VGL I2C isolation — ISO1641

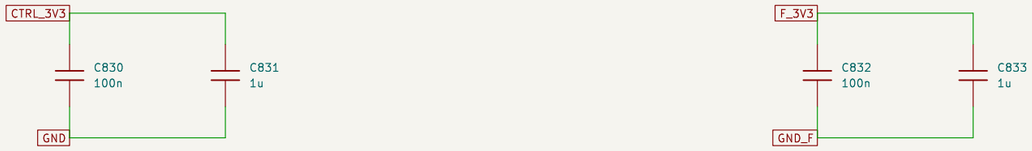
ISO1641: side 1 = system CTRL_3V3 / GND; side 2 = F_3V3 / GND_F.

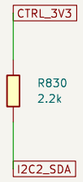
SCL direction: MCU side 1 -> floating side 2. No clock stretching.

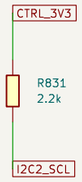
2.2k pullups on both sides are included; do not duplicate.

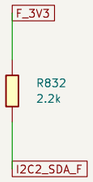
NC pins are marked individually; no shared NC net exists.

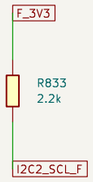












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REVIEW DRAFT — hardware validation pending

DDI Current Measurement Board

Sheet: /B8_ISO/

File: B8_ISO.kicad_sch

Title: VGL I2C isolation — ISO1641

Size: A3	Date: 2026-10-11	**Rev: 2.2-A**
KiCad E.D.A. 8.0.9		Id: 18/31

B8_PWR / VGL floating power — RFM / 3.3 V LDO

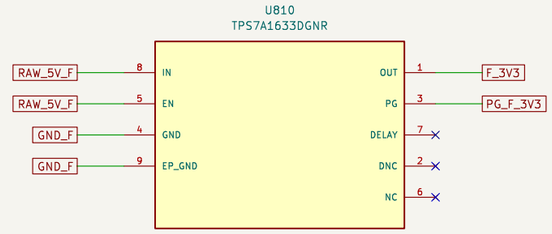
GND_F is tied ONLY to VGL_SRC through NT800; never directly to system GND.

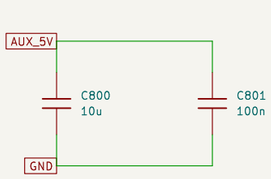
R800 = 220 ohm / >=0.5 W minimum load. F_3V3 is relative to GND_F.

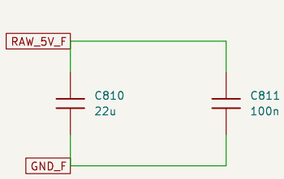
TPS7A1633: pin 2 DNC; pins 6 and 7 intentionally open.

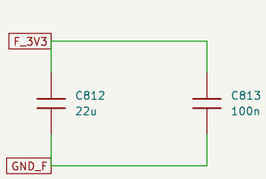
Functional floating domain; this system is not a safety-isolation design.







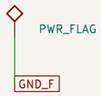












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DNP components remain in the structural netlist; their population state is stored in the symbol.

REVIEW DRAFT — hardware validation pending

DDI Current Measurement Board

Sheet: /B8_PWR/

File: B8_PWR.kicad_sch

Title: VGL floating power — RFM / 3.3 V LDO

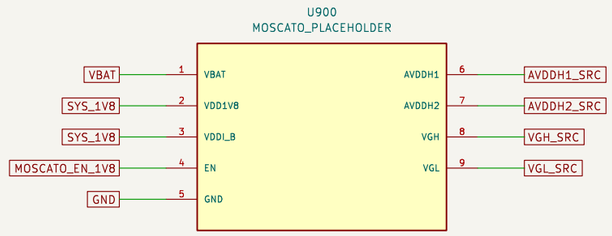
Size: A3	Date: 2026-10-11	**Rev: 2.2-A**
KiCad E.D.A. 8.0.9		Id: 19/31

B9 / MOSCATO placeholder — NOT physical pinout

STOP: this is a nine-pin LOGICAL placeholder, NOT the MOSCATO physical pinout.

Replace symbol, footprint and external support network using the actual PMIC documentation.

Do not release PCB or purchase PMIC-related passives from this placeholder.



Same-name global labels connect between sheets. Local labels connect within this sheet only.

DNP components remain in the structural netlist; their population state is stored in the symbol.

REVIEW DRAFT — hardware validation pending

DDI Current Measurement Board

Sheet: /B9/
File: B9.kicad_sch

Title: MOSCATO placeholder — NOT physical pinout

Size: A3	Date: 2026-10-11	**Rev: 2.2-A**
KiCad E.D.A. 8.0.9		Id: 20/31

B10_A / Test points 1 / power and feedback sense

All test points are connected to existing nets; no additional rail tie is introduced.

SNS pads are Kelvin probes. Never short GND_F to system GND with a grounded probe.

TP1000 VIN_12V_RAW

TP1001 VIN_12V_PROT

TP1002 PWR_RTN

TP1003 GND

TP1004 AUX_5V

TP1005 VBAT

TP1006 CTRL_3V3

TP1007 VDDIO1V2

TP1008 SYS_1V8

TP1009 VDDI_DIG_SRC

TP1010 VDDI_DIG_DUT

TP1011 VDDI_DIG_SNS_SRC

TP1012 VDDI_DIG_SNS_DUT

TP1013 VDDI_SRC

TP1014 VDDI_DUT

TP1015 VDDI_SNS_SRC

TP1016 VDDI_SNS_DUT

TP1017 VCI_SRC

TP1018 VCI_DUT

TP1019 VCI_SNS_SRC

TP1020 VCI_SNS_DUT

TP1021 AVDDH1_SRC

TP1022 AVDDH1_DUT

TP1023 AVDDH1_SNS_SRC

TP1024 AVDDH1_SNS_DUT

TP1025 AVDDH2_SRC

TP1026 AVDDH2_DUT

TP1027 AVDDH2_SNS_SRC

TP1028 AVDDH2_SNS_DUT

TP1029 VGH_SRC

TP1030 VGH_DUT

TP1031 VGH_SNS_SRC

TP1032 VGH_SNS_DUT

TP1033 VGL_SRC

TP1034 VGL_DUT

REVIEW DRAFT — hardware validation pending

DDI Current Measurement Board

Sheet: /B10_A/

File: B10_A.kicad_sch

Title: Test points 1 / power and feedback sense

Size: A3	Date: 2026-10-11	**Rev: 2.2-A**
KiCad E.D.A. 8.0.9		Id: 21/31

B10_B / Test points 2 / enable, PG, I2C and alerts

All test points are connected to existing nets; no additional rail tie is introduced.

SNS pads are Kelvin probes. Never short GND_F to system GND with a grounded probe.

TP1035 VGL_SNS_SRC

TP1036 VGL_SNS_DUT

TP1037 GND_F

TP1038 RAW_5V_F

TP1039 F_3V3

TP1040 PWR_FLT_N

TP1041 PG_AUX_5V

TP1042 PG_VBAT

TP1043 PG_CTRL_3V3

TP1044 PG_VDDI_DIG

TP1045 PG_VDDI

TP1046 PG_VCI

TP1047 PG_VDDIO1V2

TP1048 PG_SYS_1V8

TP1049 PG_F_3V3

TP1050 EN_VBAT

TP1051 EN_VDDI_DIG

TP1052 EN_VDDI

TP1053 EN_VCI

TP1054 EN_VDDIO1V2

TP1055 EN_SYS_1V8

TP1056 MOSCATO_EN_1V8

TP1057 I2C1_SDA

TP1058 I2C1_SCL

TP1059 I2C2_SDA

TP1060 I2C2_SCL

TP1061 I2C2_SDA_F

TP1062 I2C2_SCL_F

TP1063 ALERT_VDDI_DIG_N

TP1064 ALERT_VDDI_N

TP1065 ALERT_VCI_N

TP1066 ALERT_AVDDH1_N

TP1067 ALERT_AVDDH2_N

TP1068 ALERT_VGH_N

TP1069 ALERT_VGL_N

REVIEW DRAFT — hardware validation pending

DDI Current Measurement Board

Sheet: /B10_B/

File: B10_B.kicad_sch

Title: Test points 2 / enable, PG, I2C and alerts

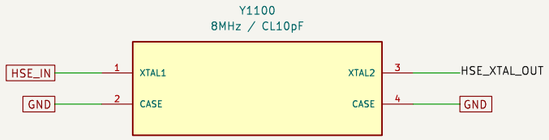
Size: A3	Date: 2026-10-11	**Rev: 2.2-A**
KiCad E.D.A. 8.0.9		Id: 22/31

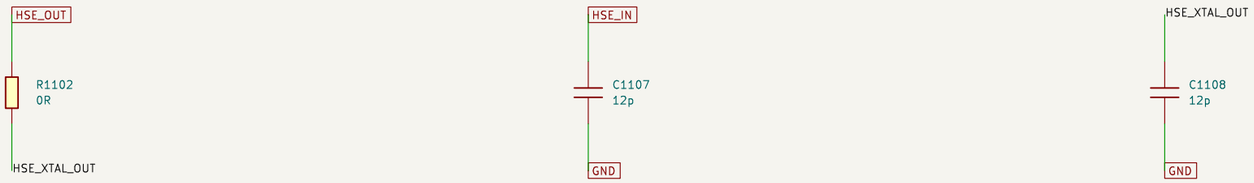
B11_CLOCK / 8 MHz HSE / calibrated timebase

8MHz crystal, CL10pF; 12pF each assumes 4pF parasitic capacitance.

Tune load/drive/startup and calibrate before claiming <=200ppm timing.

HSE failure invalidates measurement and detaches USB.





Functional pin symbols follow the Phase2-2 contract; inspect physical footprints before PCB release.

REVIEW DRAFT — hardware validation pending

DDI Current Measurement Board

Sheet: /B11_CLOCK/

File: B11_CLOCK.kicad_sch

Title: 8 MHz HSE / calibrated timebase

Size: A3	Date: 2026-10-11	**Rev: 2.2-A**
KiCad E.D.A. 8.0.9		Id: 23/31

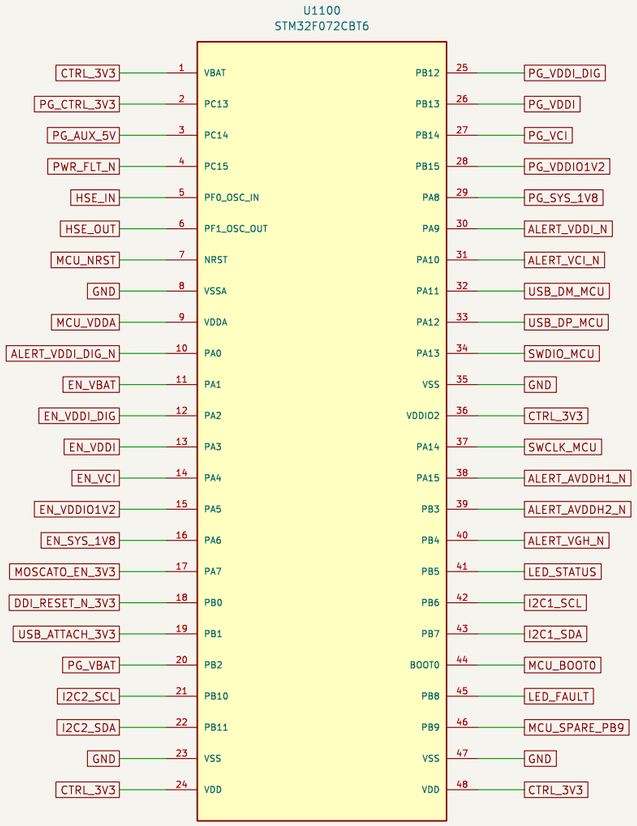
B11_CORE / STM32F072CBT6 / full 48-pin physical mapping

HSE 8MHz / PLL x6 = 48MHz. USB clock = PLL (USBSW=1).

PA0 AF2: TIM2 CH1 rising + CH2 indirect falling; 32-bit / PSC0.

EN6, MOSCATO_EN, RESET and USB_ATTACH start LOW. SWD remains enabled.

LQFP48 pin36 is VDDIO2. PC14/15 are GPIO; do not enable LSE.



Functional pin symbols follow the Phase2-2 contract; inspect physical footprints before PCB release.

REVIEW DRAFT — hardware validation pending

DDI Current Measurement Board

Sheet: /B11_CORE/
File: B11_CORE.kicad_sch

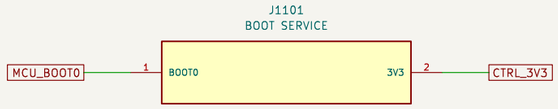
Title: STM32F072CBT6 / full 48-pin physical mapping

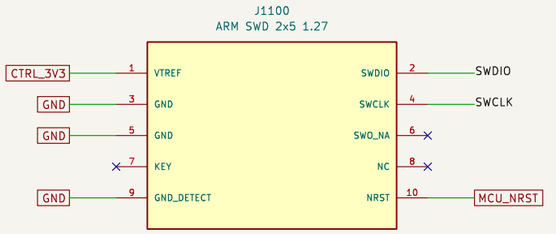
Size: A3	Date: 2026-10-11	**Rev: 2.2-A**
KiCad E.D.A. 8.0.9		Id: 24/31

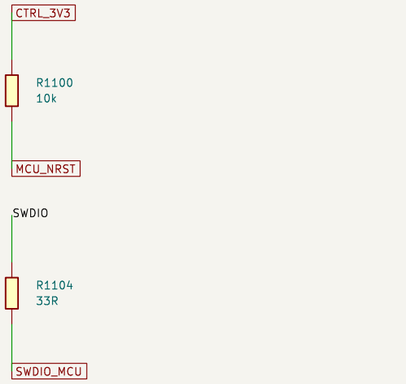
B11_DEBUG / SWD / reset / BOOT0

2x5 1.27mm SWD. VTref is sense-only: do not inject probe power.

NRST: 10k pullup / 100nF / pushbutton. BOOT0: 10k pulldown.

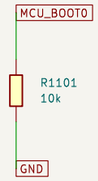


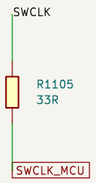












Functional pin symbols follow the Phase2-2 contract; inspect physical footprints before PCB release.

REVIEW DRAFT — hardware validation pending

DDI Current Measurement Board

Sheet: /B11_DEBUG/

File: B11_DEBUG.kicad_sch

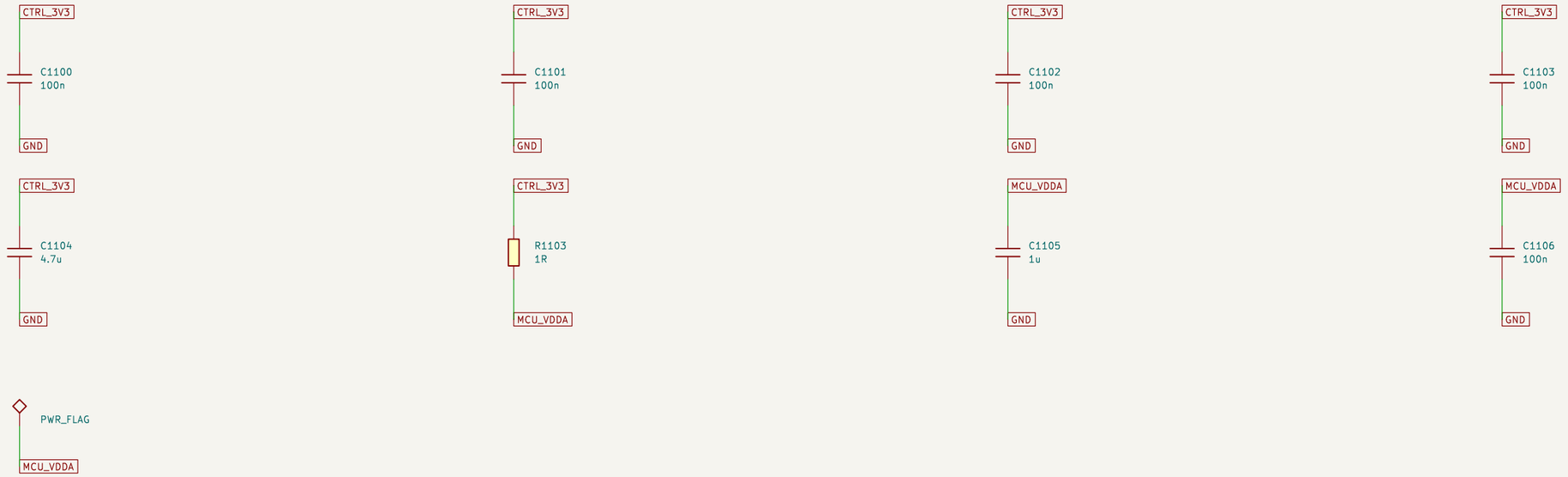
Title: SWD / reset / BOOT0

Size: A3	Date: 2026-10-11	**Rev: 2.2-A**
KiCad E.D.A. 8.0.9		Id: 25/31

B11_POWER / MCU decoupling / VDDA filter

Four 100nF caps at VDD/VDDIO2/VBAT; 4.7uF local bulk.

VDDA filter: 1 ohm + 1uF / 100nF. All grounds return to system GND.



Functional pin symbols follow the Phase2-2 contract; inspect physical footprints before PCB release.

REVIEW DRAFT — hardware validation pending

DDI Current Measurement Board

Sheet: /B11_POWER/

File: B11_POWER.kicad_sch

Title: MCU decoupling / VDDA filter

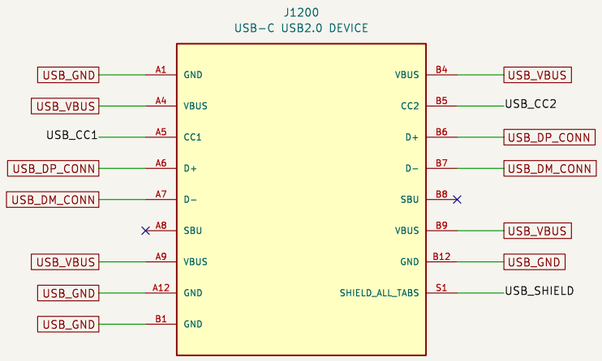
Size: A3	Date: 2026-10-11	**Rev: 2.2-A**
KiCad E.D.A. 8.0.9		Id: 26/31

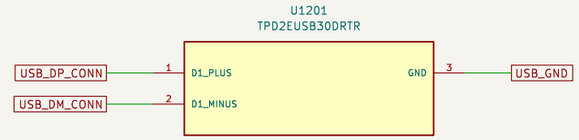
B12_CONNECTOR / USB-C host side / CC / ESD / shield

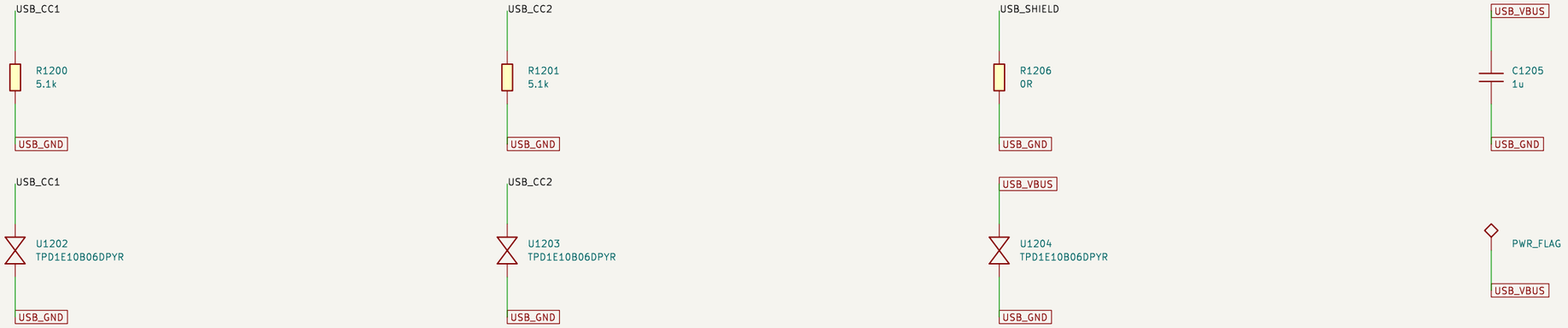
USB device: CC1 and CC2 each 5.1k to USB_GND. SBU pins NC.

USB_VBUS powers only ADuM host side. Shield connects only to USB_GND.

ESD placement and exact DRT/DPY land patterns require review.







Functional pin symbols follow the Phase2-2 contract; inspect physical footprints before PCB release.

REVIEW DRAFT — hardware validation pending

DDI Current Measurement Board

Sheet: /B12_CONNECTOR/

File: B12_CONNECTOR.kicad_sch

Title: USB-C host side / CC / ESD / shield

Size: A3	Date: 2026-10-11	**Rev: 2.2-A**
KiCad E.D.A. 8.0.9		Id: 27/31

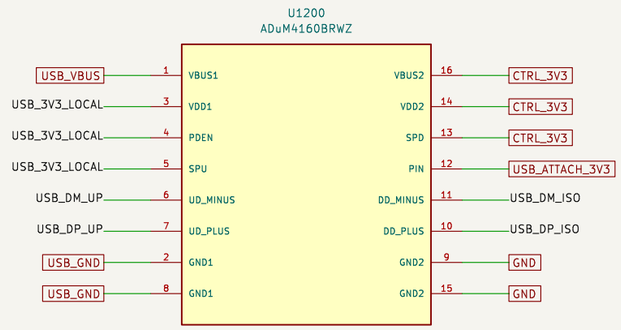
B12_ISOLATOR / ADuM4160 / full-speed isolation

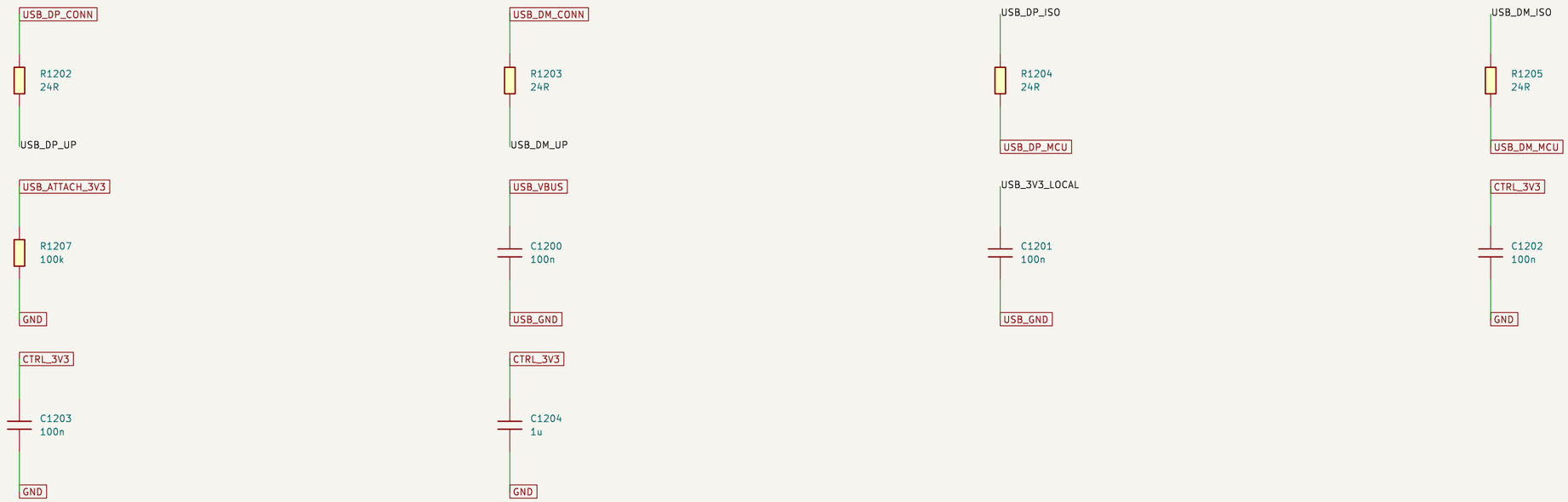
Side1: USB_VBUS / USB_GND. Side2: CTRL_3V3 / system GND.

VDD2 and VBUS2 both 3.3V. SPU/SPD high = full speed.

24 ohm resistors on both D+/D- pairs. PIN stays low until USB ready.

Never bridge USB_GND to GND/GND_F. Test suspend and all power combinations.





Functional pin symbols follow the Phase2-2 contract; inspect physical footprints before PCB release.

REVIEW DRAFT — hardware validation pending

DDI Current Measurement Board

Sheet: /B12_ISOLATOR/

File: B12_ISOLATOR.kicad_sch

Title: ADuM4160 / full-speed isolation

Size: A3	Date: 2026-10-11	**Rev: 2.2-A**
KiCad E.D.A. 8.0.9		Id: 28/31

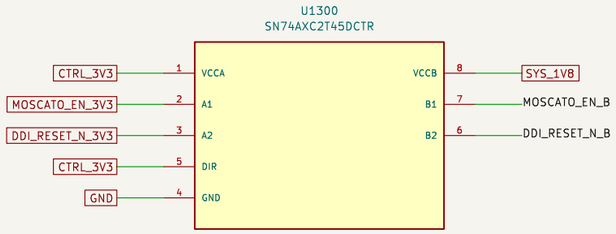
B13 / 3.3 V to 1.8 V / MOSCATO EN and DDI reset

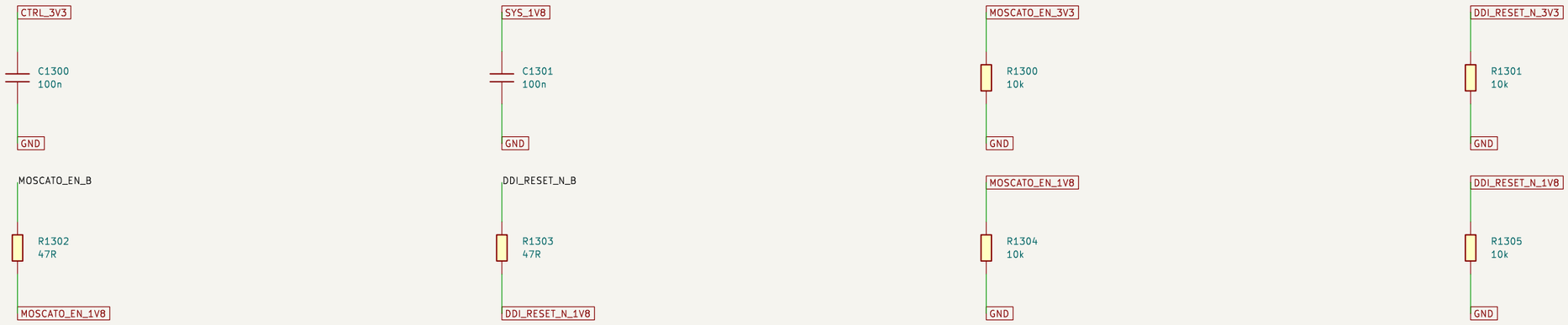
DIR high = A to B. VCCA 3.3V / VCCB SYS_1V8. No OE pin.

External 10k pulldowns hold MOS_EN low and RESET asserted at boot.

Below 0.1V on either supply: high-Z; Ioff budget 8uA -> 80mV on 10k.

Partial-power ramp and external DUT backpower require bench verification.





Functional pin symbols follow the Phase2-2 contract; inspect physical footprints before PCB release.

REVIEW DRAFT — hardware validation pending

DDI Current Measurement Board

Sheet: /B13/

File: B13.kicad_sch

Title: 3.3 V to 1.8 V / MOSCATO EN and DDI reset

Size: A3	Date: 2026-10-11	**Rev: 2.2-A**
KiCad E.D.A. 8.0.9		Id: 29/31

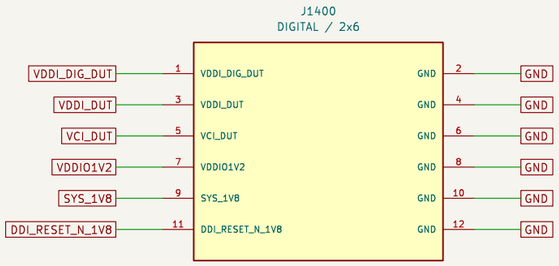
B14 / DUT connectors / adjacent ground returns

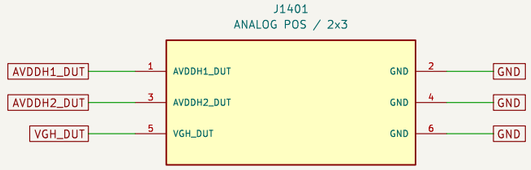
Each power/control signal has adjacent system GND return.

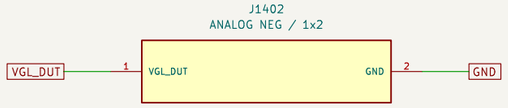
Keep VGL connector separate from positive analog connector (initial 10mm).

GND_F is NOT system ground and is NOT routed to these headers.

Unkeyed 2.54mm connectors: verify harness polarity before power-on.







Functional pin symbols follow the Phase2-2 contract; inspect physical footprints before PCB release.

REVIEW DRAFT — hardware validation pending

DDI Current Measurement Board

Sheet: /B14/

File: B14.kicad_sch

Title: DUT connectors / adjacent ground returns

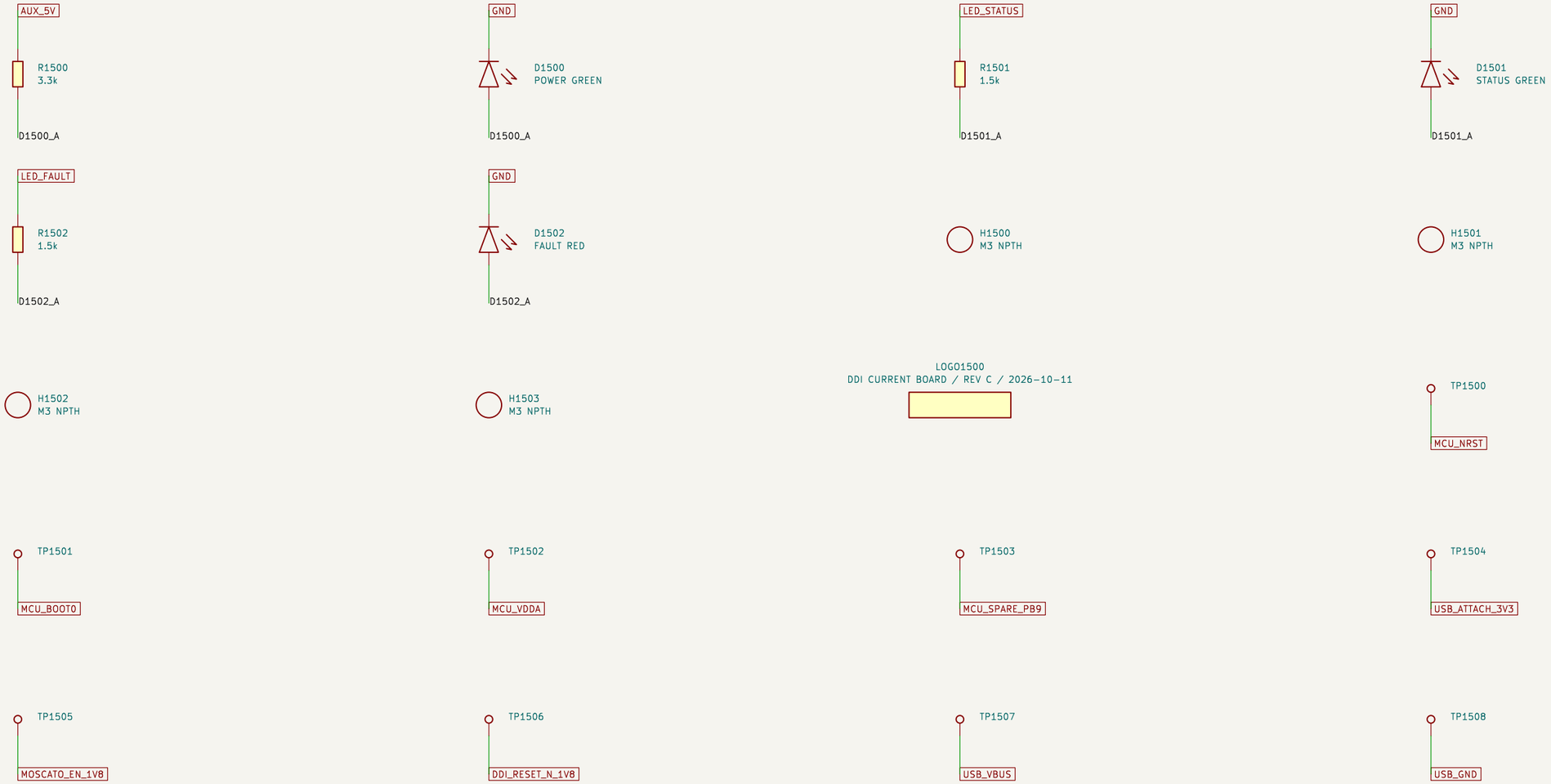
Size: A3	Date: 2026-10-11	**Rev: 2.2-A**
KiCad E.D.A. 8.0.9		Id: 30/31

B15 / LEDs / mechanical / silkscreen / debug pads

LED power is not drawn from measured rails.

M3 holes: copper keepout >=2mm from hole edge; keep isolation barriers clear.

Silk: REV C / serial / calibration ID / connector polarity / ground domains.



Functional pin symbols follow the Phase2-2 contract; inspect physical footprints before PCB release.

REVIEW DRAFT — hardware validation pending

DDI Current Measurement Board

Sheet: /B15/

File: B15.kicad_sch

Title: LEDs / mechanical / silkscreen / debug pads

Size: A3	Date: 2026-10-11	**Rev: 2.2-A**
KiCad E.D.A. 8.0.9		Id: 31/31